

Transformer-based deep learning model and video dataset for unsafe action identification in construction projects

Meng Yang^{a,b}, Chengke Wu^a, Yuanjun Guo^{a,b,e}, Rui Jiang^a, Feixiang Zhou^c, Jianlin Zhang^d, Zhile Yang^{a,b,e,*}

^a Shenzhen Institute of Advanced Technology, Chinese Academy of Sciences, Shenzhen 518055, China

^b University of Chinese Academy of Sciences, Beijing 100049, China

^c School of Computing and Mathematical Sciences, University of Leicester, Leicester LE1 7RH, UK

^d China Construction Science and Technology Group Cooperation, Room 703, G3 Building, TCL international E-city Nanshan, Shenzhen, China

^e Guangdong Institute of Carbon Neutrality, Building 41, Huangshaping Innovation Park, Phase I, Guanshao City, Wujiang District, Shaoguan, China

ARTICLE INFO

Keywords:

Action recognition
Construction safety
Transformer
Deep learning

ABSTRACT

A large proportion of construction accidents are caused by unintentional and unsafe actions and behaviors. It is of significant difficulties and ineffectiveness to monitor unsafe behaviors using conventional manual supervision due to the complex and dynamic working conditions on construction sites. Recently, surveillance videos and computer vision (CV) techniques have been increasingly adopted to automatically identify risky behaviors. However, the challenge remains that spatial and temporal features in video clips cannot be effectively captured and fused by current CV models. To address this challenge, this paper describes a deep learning model named Spatial Temporal Relation Transformer (STR-Transformer), where spatial and temporal features of work behaviors are simultaneously extracted in paralleling video streams and then fused by a specially designed module. To verify the effectiveness of the STR-Transformer, a customized dataset is developed, including seven categories of construction worker behaviors and 1595 video clips. In numerical experiments and case studies, the STR-Transformer achieves an average precision of 88.7%, 4.0% higher than the baseline model. The STR-Transformer enables more accurate and reliable automatic safety surveillance on construction projects, and is expected to reduce accident rates and management costs. Moreover, the performance of STR-Transformer relies on efficient feature integration, which may inspire future studies to identify, extract, and fuse richer features when applying CV-based deep learning models in construction management.

1. Introduction

Safety has long been a serious challenge in the construction industry. According to Ministry of Emergency Management of China, the death toll in the Chinese construction industry was 1752 during the first half of 2018 [1]. In USA, an average of 5136 workers died while working on construction sites annually from 2016 to 2020 [2]. Also note that the construction accident rate remains high in recent years despite continuously released safety regulations, manuals, and standards [3,4]. Construction accidents are mainly caused by unsafe behaviors of construction workers. Jiang et al. [5] and Heinrich et al. [6] reported that approximately 80%–88% of all accidents occurring during construction were caused by unsafe human behaviors. On construction sites, most accidents are visually identical. Casualty and losses

could be avoided if unsafe behaviors during the construction processes could be detected in time and addressed in an appropriate manner.

Construction workers' behavior monitoring mainly relied on human supervision in the past, but this approach has many limitations. On the one hand, manual identification of unsafe movements is inaccurate and delayed due to the large number of workers and complex conditions on large construction sites [7]. On the other hand, supervision tasks require additional safety engineers, which will inevitably increase project management costs [8]. In recent years, with the development of smart construction technology, some studies in sensor monitoring field have to some extent solved these problems. Various sensors help collect information related to worker movements by recognizing specific signals [9–11]. However, these wearable sensors used to extract signals

* Corresponding author.

E-mail addresses: yangmeng@siat.ac.cn (M. Yang), ck.wu@siat.ac.cn (C. Wu), yj.guo@siat.ac.cn (Y. Guo), rui.jiang@siat.ac.cn (R. Jiang), fz64@leicester.ac.uk (F. Zhou), zhang-jl@cscec.com (J. Zhang), zl.yang@siat.ac.cn (Z. Yang).

<https://doi.org/10.1016/j.autcon.2022.104703>

Received 16 June 2022; Received in revised form 9 November 2022; Accepted 3 December 2022

Available online 14 December 2022

0926-5805/© 2022 Elsevier B.V. All rights reserved.

usually need to be attached to the body while working, potentially leading to physical interference with the worker's movements [12].

To this end, surveillance cameras are increasingly deployed on construction sites, which enables remote automatic monitoring of project elements, e.g., structures, machines, materials, as well as workers. Thousands of real-time images and videos are collected every day, covering the entire construction site. When efficiently handled, these valuable data can enable Artificial Intelligence (AI) methods, thereby largely reducing human effort in safety management and reduce project costs [13]. Many studies have therefore applied computer vision (CV) to monitoring the wearing status of personal protective equipment (PPE) on construction sites, such as hard hats and vests [14–16]. Meanwhile, some research uses CV-based methods to avoid workers falling from heights (FFH) [17,18]. However, these studies often use image-based methods, detecting objects in discrete images rather than recognizing consecutive actions. As a result, complex behaviors that require temporal information in consecutive video frames cannot be accurately identified to determine whether they are risky.

This issue has been addressed in several recent studies in the field of action recognition [19,20], which aims to analyze and identify actions that occur within a short period in videos. Such methods have also been applied in studies attempting to recognize worker actions in construction projects. However, these studies all have one or several of the following limitations: (1) Most skeleton-based studies [21–23] extract and classify skeleton features instead of directly recognizing worker actions from videos, which ignore important visual features; (2) Several studies [24–27] have identified worker movements directly from videos. However, much attention has been paid to spatial features within each area of the frame. The connections between different areas of the whole frame were ignored since the convolution kernel cannot model the dependencies beyond the receptive field; (3) They focus on the feature extraction ability of the model without fully considering the fusion of spatial and temporal features in video clips. Improper fusion strategy may probably lead to loss of information, reduction of feature representation ability, and a decrease in model precision. Besides, most of the existing datasets [15,28] are private and/or are proposed for only a few specific behaviors (e.g., climbing and smoking). As a result, existing models within the same publicly available datasets are difficult, if possible, to be fully compared; and existing datasets can hardly extend well to subsequent studies of unsafe behaviors.

To address these limitations of current studies, a Transformer-based model, i.e., Spatial Temporal Relation Transformer (STR-Transformer), is proposed in this paper to recognize unsafe actions of construction workers from videos. A Spatial Feature Encoder (SFE) based on Transformer is developed to extract spatial features in each frame, where a unique self-attention mechanism is employed to aggregate information about scenes and objects in each frame. An additional 3D convolutional neural network (3D CNN) is applied as a Temporal Feature Encoder (TFE) to extract temporal information about the movement of objects from the video. The SFE and TFE will process different types of inputs to generate multimodal features, such as RGB frames and RGB difference frames, and can effectively perform both spatial extraction enabled by the Transformer and temporal extraction enabled by 3D CNN. In addition, the fusion of spatial and temporal features largely affects the final prediction performance. Therefore, a Relative Position Feature Fusion Model (RPFFM) is proposed, which enhances spatial features and position correlations in each frame, while producing fused temporal-spatial features for final prediction. Furthermore, a novel dataset, named Construction Meta Action (CMA) is built, containing 1595 video clips and seven classes of worker actions. These videos are collected from Youtube considering data availability, and the actions in the CMA dataset represent daily behaviors of workers and can be reused in many scenarios. In summary, this paper makes the following contributions:

- A Transformer-based model named STR-Transformer is developed, which can well extract the spatial-temporal features for the construction sites unsafe action recognition task. Experimental results prove that the proposed model outperforms the state-of-the-art (SOTA) methods on a standard dataset developed in this study.
- A RPFFM is proposed, which can be added to various deep learning models to enhance their fusion ability for multimodal features and improve recognition precision.
- A novel dataset (i.e., CMA) is proposed for action recognition application on construction sites. This dataset includes seven common worker behaviors, while also allowing flexible expansion.

The remaining of this paper is organized as follows. Section 2 provides a review of the development of action recognition models on construction sites, including the current work of Transformer and its applications. Section 3 describes the proposed Transformer-based model in detail and Section 4 presents the dataset of worker actions. In Section 5, details of experiment implementation are presented and the experimental research results are analyzed and compared. Findings and limitations are discussed in Sections 6 and 7 presents conclusions.

2. Related work

2.1. Detecting unsafe behavior through signals or images

Extensive efforts have been made to monitor unsafe behaviors through various signals collected from workers on construction sites. Li et al. [12] combined convolutional neural network (CNN) with long short-term memory (LSTM) [29] to predict unsafe movements from acceleration signals. Bangaru et al. [11] identified activities of builders through electromyography and inertial measurement unit. Antwi-Afari et al. [30] used wearable equipment to identify awkward working postures. Jung et al. [10] proposed a human activity classification model based on a sound recognition method. Lee et al. [31] developed a new audio-based event detection system for automated construction safety surveillance. However, some wearable sensors used to extract signals can potentially lead to physical interference in the worker's movements. Due to the low dimensionality of the signals and limited information it carries, the model cannot fully learn the features of the signals. In addition, the specific signal required for the research can be easily interfered by other signals on construction sites, resulting in extra time for subsequent data cleaning.

Failing to wear PPE on construction sites is an unsafe behavior that can result in various accidents such as FFH. Many injuries and deaths could be avoided if workers have worn proper PPE. Several studies have attempted to use CV techniques to automatically monitor the appropriate use of PPE in images. For example, Fang et al. [28,32] proposed an automated inspection method to check the appropriate usage of PPE for working at height. Nath et al. [14] detected whether workers were wearing hard hats and vests through deep learning methods. Xiong and Tang [15] proposed a pose-guided anchoring framework to address the challenges of multi-class PPE detection in a workspace. Yang et al. [16] applied a deep learning model to detect whether workers were wearing helmets. Chian et al. [17] developed two CV-based detection methods to automatically detect missing barricades. Fang et al. [18] developed a visual model to identify construction workers traversing on the structure and determine the relative positioning of the workers and the structural support. These image-based methods focus more on recognizing and localizing objects in images rather than recognizing actions. For example, unsafe actions with easily identified objects (e.g., not wearing a helmet) are usually distinguished by determining whether the object (e.g., the helmet) can be identified from an image. However, complex unsafe behaviors usually last for a period of time, making them difficult to be accurately identified by these image-based methods. For example, "Fall Down" requires temporal information in consecutive video frames to determine whether they are dangerous because the information from only one frame can be misleading.

2.2. Detecting unsafe behavior through videos

To address limitations in detecting unsafe behaviors from signals or images, researchers have attempted to identify unsafe behaviors by understanding videos. Recently, deep learning based approaches have been widely used in human action recognition, animal behavior analysis [33–35] and smart construction [36,37]. Among them, two-stream-based methods and 3D-CNN-based methods are two well-known deep learning models for action recognition. Two-stream-based methods [20,38,39] usually have two different networks to handle different types of inputs, such as RGB images, RGB difference images and Optical Flow. The two networks are trained separately and features extracted by them will be fused in the middle or at the end to get a final prediction. This method can help obtain diverse features with very high precision as long as kinds of features are well integrated. 3D-CNN-based methods [19,40,41] use 3D convolutional kernels to extract both spatial and temporal features in one operation from a video clip, which extend the 2D convolutional kernels with an extra dimension. Due to the heavy computational burden of 3D CNN, some studies [42–44] proposed new 3D convolution operations to reduce the computation cost. Others [45–48] also designed a variety of modules that could be directly added to various networks to improve the ability of feature extraction of the original networks.

With the development of deep learning models for action recognition tasks, many researchers have introduced these models into construction engineering to detect unsafe actions of workers or construction equipment. Existing studies often identify abnormal behaviors by modeling worker skeleton data. For example, Han and Lee [49] employed a stereo camera to extract 2D skeleton points and reconstructed 3D human skeleton point coordinates to monitor unsafe behaviors. Roberts et al. [21] built a skeleton-based model to estimate and track worker poses and outputs action labels. Li and Li [8] solved the occlusion problem of construction workers' action recognition with the skeleton-based model. Ding et al. [22] proposed a temporal segment graph convolutional network to model skeleton data and fuse features in the temporal dimension for action recognition. Subedi and Pradhananga [23] employed a depth sensor camera to get the skeletal positional data and then proposed a machine learning method to identify the actions. These skeleton-based models are two-stage models that first extract skeleton information from the videos, and then classify worker actions through the skeleton rather than video frames. The performance of the first-stage skeleton extraction model has a great impact on the final result, and these models tend to be time-consuming because their two stages cannot be parallelized.

As such, researchers start to identify worker behaviors directly from videos. Luo et al. [27] propose a two-stream network and a new fusion strategy to recognize workers' activities in site surveillance videos. However, the fusion strategy used to merge the results of two streams is still relatively simple, which may lead to information loss. Ding et al. [25] proposed a hybrid deep learning model based on LSTM to automatically extract and recognize workers' actions of climbing and dismounting from a ladder. This method obtains 25 feature vectors of $\in \mathbb{R}^{2048}$ from a video and feeds them into the LSTM model, making it time-consuming. Luo et al. [24] introduced a discriminative model that combined deep activity features and contextual information to improve its ability of action recognition. Yang et al. [50] conducted a study using Dense Trajectories [51] to recognize workers' actions. At the same time, it is also very important to monitor the operation status of construction equipment. Jung et al. [26] applied a 3D-CNN-based method to recognize the action of construction equipment such as excavators and trucks from videos. Kim et al. [52] employed CNN and Double-layer LSTM to identify the working status of excavators. Bügler et al. [53] integrated two vision-based sensing methods to generate productivity information regarding excavation operations. In conclusion, a large number of articles have been published on the identification of construction worker behaviors. These studies have

greatly facilitated the introduction of CV techniques into architectural applications. However, with the advent of the digital twin era, the amount of data is growing exponentially, and unsafe behaviors to be identified are getting more and more complex. Current methods, which are based on CNN and LSTM, are difficult to meet such demands due to the following shortcomings: (1) The convolution kernel cannot model dependencies beyond the receptive field, resulting in the inability to learn long-range spatial-temporal information well. (2) LSTM has a long-term memory function and has certain advantages when dealing with sequence modeling problems. However, when the temporal sequence is too long or the number of LSTM layers is very deep, the computation can be very time-consuming due to the failure to perform parallel computing across layers.

2.3. Transformer and applications

To address the issues of CNN and LSTM, Transformer [54] stands out. Transformer is originally proposed to solve natural language processing (NLP) tasks. Due to its computational efficiency and scalability, many studies have introduced Transformer to CV tasks [55–58]. Transformer can apply a self-attention mechanism to capture local and global long-range dependencies, which is far beyond the receptive field of traditional convolutional filters. Moreover, Transformer can be computed in parallel, which can greatly reduce the computation time compared to LSTM. Ensuring Transformer for better extracting features of images or videos is a key challenge for its application in CV. Raghu et al. [59] explored the differences between Transformer and CNN to find out a way Transformer could solve the CV tasks. Dosovitskiy et al. [55] applied the standard Transformer directly to images. By splitting the images into patches and providing a linear sequence embedding of these patches, which are treated the same way as tokens in the NLP application, the lowest level of modifications was made to ensure its best performance. Arnab et al. [60] developed four pure-Transformer models that can efficiently encode and process massive spatial and temporal features of the videos. Bertasius et al. [61] proposed a spatially and temporally based self-attention method that adapted the standard Transformer architecture to videos by directly learning spatiotemporal features from a sequence of frame-level patches. Liu et al. [62] proposed a shifted window mechanism to calculate the attention in the spatiotemporal domain, which can achieve better results in reducing the amount of model parameters. Fan et al. [63] connected the seminal idea of multiscale feature hierarchies with Transformer models. Yan et al. [64] used separate encoders to represent different views of the input videos and fused information across views through lateral connections. Tolstikhin et al. [65] replaced the attention in Transformer with multi-layer perceptions to verify the necessity of the attention mechanism.

Due to the excellent performance of Transformer in various tasks, researchers have recently applied Transformer to the construction industry. Shamsabadi et al. [66] employed Vision Transformer (ViT) for crack detection in civil infrastructure. Wang and Su [67] proposed a novel Transformer-based crack segmentation model. Zhou et al. [68] applied Transformer to a point cloud classification task. Moon et al. [69] used Bidirectional Encoder Representation from Transformers (BERT) [70] to automatically detect contractual risk clauses from construction specifications. In these studies, Transformer has shown stronger feature extraction capabilities than CNN and LSTM. However, to the best of the authors' knowledge, no research has been done on applying Transformer to automatically identifying unsafe worker behaviors on construction sites. This study introduces Transformer to the task of recognizing unsafe actions of construction workers, and the performance is proven to be better than CNN-based and LSTM-based models.

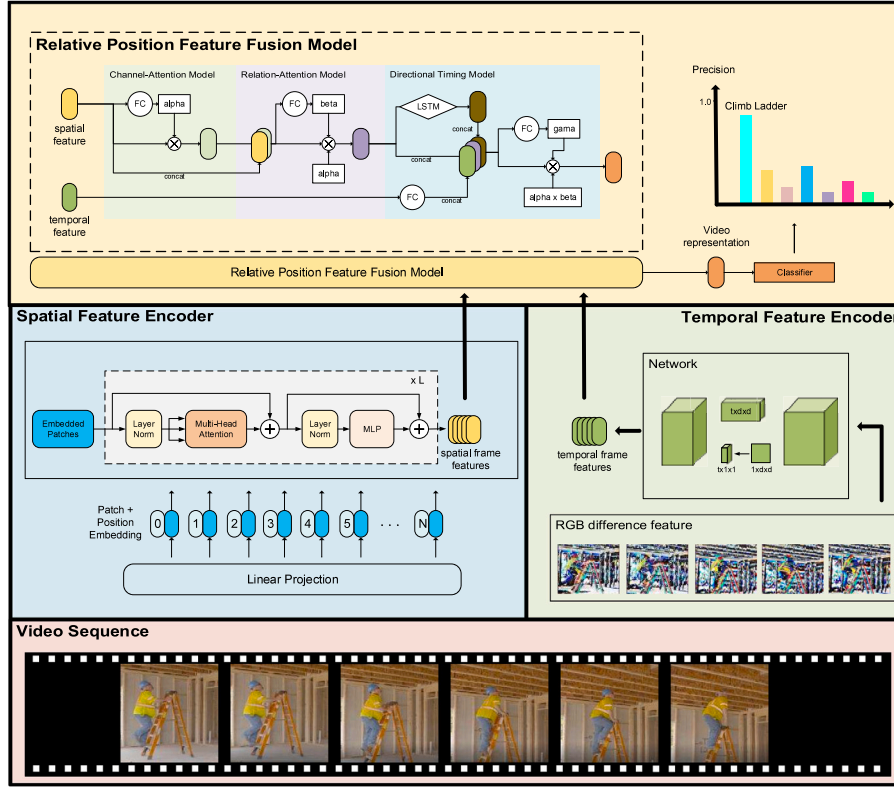


Fig. 1. Main architecture of STR-Transformer.

3. Method

In this section, the proposed SFE, TFE, RPFFM are introduced in detail. The overall architecture of the STR-Transformer is shown in Fig. 1. Two well-known deep learning models, Transformer and 3D CNN, are applied to encode spatial and temporal features, respectively. Two types of video data, RGB and RGB difference video clips, are fed into the SFE and TFE separately to generate multimodal features. Finally, a RPFFM is adopted to fuse and enhance the two types of features for the final prediction.

Currently, mainstream Transformer-based action recognition models usually add a temporal attention module to the spatial attention module, which changes the original architecture of the Transformer and makes the pre-trained weights unable to be applied to the new model. Inspired by two-stream network, in the STR-Transformer, an additional TFE based on 3D CNN is added to extract temporal features without changing the original architecture. Moreover, a RPFFM consisting of three sub-models is developed to enhance the spatial and temporal features based on channel importance and position relationship, thereby improving the effectiveness of feature fusion. The advantage of the model design is that Transformer models pre-trained on other large datasets can be directly transferred to the task in hand, thereby avoiding training a Transformer from scratch and saving enormous computational costs.

3.1. Feature extraction

Videos can be naturally decomposed into spatial and temporal features. The spatial domain is formed by individual frames and carries information about scenes and objects depicted in the video. On the other hand, the temporal domain covers different frames and carries information of the movements and changes of movements of the observer (i.e., the camera) and objects [38]. Both spatial and temporal features play essential roles in video understanding.

3.1.1. Spatial feature encoder

The Transformer [54,55,71] is applied as the SFE to extract spatial features. Specifically, k frames are sampled from each video following the sampling method in Temporal Segment Network (TSN) [20], producing the input sequence $T = (T_1, T_2, \dots, T_k)$. Each frame $T_i \in \mathbb{R}^{H \times W \times C}$ will be reshaped into a sequence of flattened 2D patches $x_i \in \mathbb{R}^{N \times (P^2 \times C)}$, where H, W, C is the height, width, and number of channels of original frame, respectively; P is the width/height of each patch, respectively; $N = HW/P^2$ is the number of patches. In order to satisfy the input dimension D of the transformer, the patches are flattened and reshaped to D with a trainable linear projection $E \in \mathbb{R}^{(P^2 \times C) \times D}$ following the equation below:

$$z_0 = [x_{class}; x_1^1 E; x_1^2 E; \dots; x_1^N E] + E_{pos} \quad (1)$$

where z_0 is the patch embedding, which is the output of Eq. (1); x_{class} is learnable classification embedding and $E_{pos} \in \mathbb{R}^{(N+1) \times D}$ represents the standard learnable position embedding [55]. Then, z_0 is fed into the Transformer encoder, producing the output S_i as follows:

$$z'_l = MSA(LN(z_{l-1})) + z_{l-1} \quad l = 1 \dots L \quad (2)$$

$$z_l = MLP(LN(z'_l)) + z'_l \quad l = 1 \dots L \quad (3)$$

$$S_i = LN(z_L^0) \quad (4)$$

where LN is the layer normalization [72]; L represents the number of Transformer Encoder; MSA represents the Multi-head self-attention (MSA) mechanism which is defined as follows:

$$[Q, K, V] = zU_{qkv} \quad (5)$$

$$SA(Q, K, V) = \text{SoftMax}\left(\frac{QK^T}{\sqrt{d}}\right)V \quad (6)$$

where $Q, K, V \in \mathbb{R}^{(N+1) \times d}$ indicate the query, key, and value matrices; z is the input of MSA , and $U_{qkv} \in \mathbb{R}^{D \times 3d}$ is the learnable parameters to calculate Q, K, V ; d represents the dimension of query, key, and value; SA is the standard self-attention mechanism; $SoftMax$ represents the softmax function. The MSA performs self-attention operations n_{msa} times and then concatenates the results as follows:

$$MSA(z) = [SA_1(z); SA_2(z); \dots; SA_{n_{msa}}(z)]U_{msa} \quad U_{msa} \in \mathbb{R}^{D \times D} \quad (7)$$

where U_{msa} is the learnable parameters in the Multi-head self-attention; MLP represents the Multi-Layer Perception defined as follows:

$$MLP(x) = FC(Gelu(FC(x))) \quad (8)$$

where x is the input vector; FC represents the fully connected (FC) layer, and $Gelu$ represent the GELU function. All k frames in the input video will go through the SFE, and the final output is $S = (S_1, S_2, \dots, S_k)$.

3.1.2. Temporal feature encoder

Considering the widely known capacity of the 3D-CNN-based model in terms of processing consecutive images, it is applied as the TFE to extract temporal features in videos [19]. The RGB difference image between two consecutive frames can effectively reveal the appearance change, therefore facilitating identification of motion salient regions [20]. Thus, the proposed TFE takes RGB difference image sequence as the input. First, for the input RGB image sequence $T = (T_1, T_2, \dots, T_k)$, the RGB difference sequence $Diff$ is generated as follows:

$$Diff_i = 0 \quad i = 1 \quad (9)$$

$$Diff_i = T_i - T_{i-1} \quad i = 2 \dots k \quad (10)$$

Then $Diff \in \mathbb{R}^{k \times C \times H \times W}$ will go through the 3D-CNN-based model as follows:

$$M = Relu(BN(Conv3D(Diff))) \quad (11)$$

where $Conv3D$ represents 3D convolution operations; BN is the batch normalization (BatchNorm) operation; $Relu$ is the relu activation function; M is the output of TFE.

3.2. Relative position feature fusion model

A RPFPM is developed to fuse and enhance the spatial features S and temporal features M . The fusing model produces a fixed-dimension feature representation as the output for final action recognition prediction. The model is designed based on the previous studies [73–76], which consists of three sub-models: Channel-Attention Model (CAM), Relation-Attention Model (RAM), and Directional Timing Model (DTM).

3.2.1. Channel-attention model

It is believed that for action recognition, using representative and distinct features for final prediction usually performs better than using a larger number of unimportant features [73,74]. Therefore, a fully connected layer FC and a sigmoid function σ are adopted to filter and select representative features. The mathematical representation is represented in Eq. (12), where α_i is the self-attention weight of the i th channel.

$$\alpha_i = \sigma(FC(S_i)) \quad (12)$$

Then, following the method in Meng et al. [74], the self-attention weight is combined with spatial features to generate the channel enhancement feature, i.e., Θ_{alpha} . In other words, the CAM is expected to identify important channels of input features so that subsequent models can pay more attention to salient and representative features.

$$\Theta_{alpha} = \frac{\sum_{i=1}^k \alpha_i S_i}{\sum_{i=1}^k \alpha_i} \quad (13)$$

3.2.2. Relation-attention model

In STR-Transformer, the spatial features extracted by SFE are treated as local features because they contain the information in each frame, and the channel enhancement feature generated by CAM is regarded as the global feature because it inherently contains the contents of the whole video. The relation between local features and global feature represents the correlation of information carried by each frame and the entire video. Learning such relation can increase the precision and reliability of action recognition [74,76]. The two features are concatenated as $[S_i; \Theta_{alpha}]$ and then fed into a FC layer to evaluate the relation-attention weight β as follows:

$$\beta_i = \sigma(FC([S_i; \Theta_{alpha}])) \quad (14)$$

where β_i is the relation-attention weight of the i th feature, and the σ donates the sigmoid function. As such, with the self-attention and relation-attention weights, the model will generate two new features Φ_i and Θ_{beta} , which are taken by the third sub-model, i.e., the DTM.

$$\Phi_i = \alpha_i \beta_i [S_i; \Theta_{alpha}] \quad (15)$$

$$\Theta_{beta} = \frac{\sum_{i=1}^k \Phi_i}{\sum_{i=1}^k \alpha_i \beta_i} \quad (16)$$

3.2.3. Directional timing model

The CAM and RAM focus more on the features in each frame. However, the time information across frames is also important. Thus, a LSTM is applied to learn the motion changes in videos and enhance the representation of temporal features as follows:

$$O, h_t, c_t = LSTM(\Phi, h_{t-1}, c_{t-1}) \quad (17)$$

where O is the output of LSTM; h, c is the learned parameters of LSTM. Then, the three features are stacked and fed into another FC layer to produce the Timing-Attention weight as follows:

$$\gamma_i = \sigma(FC([O_i; M; \Theta_{beta}])) \quad (18)$$

where σ denotes the sigmoid function. The DTM aggregates all the features above to generate Θ_{gamma} , which will be the input of the final classifier and softmax function to produce the prediction Y of the action in the current video.

$$\Theta_{gamma} = \frac{\sum_{i=1}^k \alpha_i \beta_i \gamma_i [O_i; M; \Theta_{beta}]}{\sum_{i=1}^k \alpha_i \beta_i \gamma_i} \quad (19)$$

$$Y = SoftMax(FC(\Theta_{gamma})) \quad (20)$$

4. Dataset

4.1. An overview of the CMA dataset

The CMA dataset focuses on seven classes of basic worker actions on construction sites, which are divided into three groups: *Single-Person Actions*, e.g. “Stand”, “Smoke”, “Walk” and “Fall Down”, means that these actions can be independently finished by a single person; *Person–Person Actions*, e.g. “Talk”, refers to actions requiring the cooperation of multiple people; and *Person–Object Actions*, e.g. “Climb Ladder” and “Wear Helmet”, indicates that these actions usually require human interaction with an object. Definitions of these action classes are shown in Table 1 and all defined actions are potentially dangerous on construction sites. For example, the actions of “Talk” and “Smoke” are related to distraction of workers which is the cause of unsafe behaviors and safety performance degradation [77]. Detecting improper use of PPE is crucial for promoting safety management in construction workplaces [15] and “Wear Helmet” is closely related to this issue. During construction work, the gestures of workers such as “Stand”, “Walk” and “Fall Down” are related to slips, trips and falls which have been recognized as significant sources of injuries on construction sites [78]. Moreover,

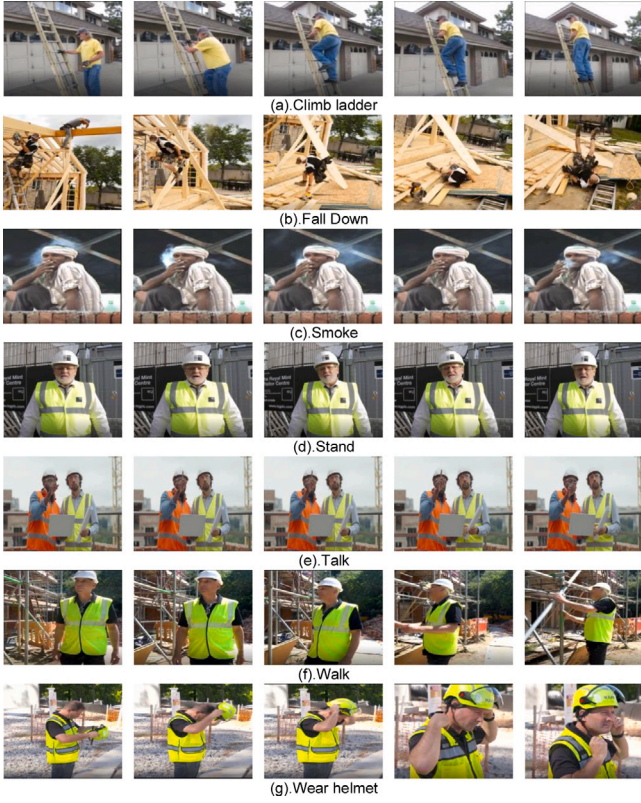


Fig. 2. Sample frames for seven action classes of CMA dataset.



Fig. 3. Examples of video clips after data augmentation.

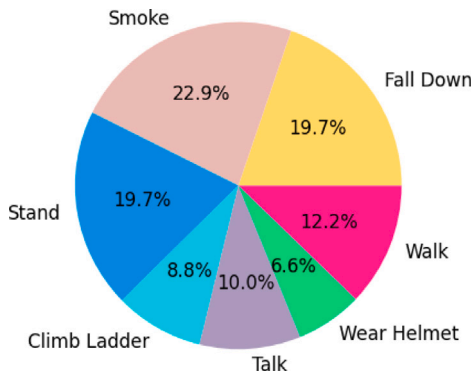


Fig. 4. Data proportion for seven class of CMA dataset.

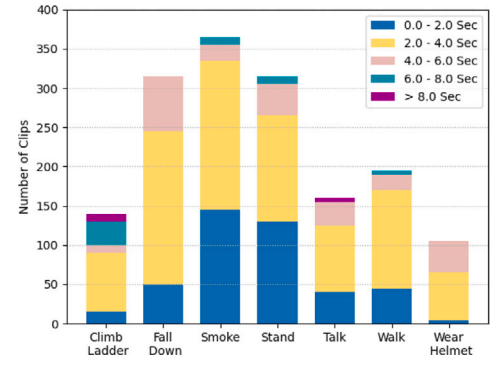


Fig. 5. Number of clips per action class. The distribution of clip durations is illustrated by colors.

according to Fang et al. [18], the class “Climb Ladder” is related to Falls From Height which constitutes the highest portion of construction accidents. Thus, these actions should be added into CMA. Fig. 2 shows sample frames for the seven action classes in the CMA dataset. It can be seen from the figure that some actions such as “Stand” and “Walk” require temporal information to distinguish. Some other actions need to be identified by distinguishing objects in the video, such as “Climb Ladder” and “Wear Helmet”.

Existing datasets consisting of signal data [12,77] and image data [14,15] usually do not have complete visual features or spatial-temporal features. Other datasets, although contain video data, focused on one specific risky behavior (e.g., FFH or not wearing PPE) in each study [8,25,27,32], but may not be suitable for other construction scenarios. Based on video data, the CMA dataset preserves original visual features and selects seven worker behaviors that are commonly ignored by existing datasets. It is also expected to be widely used in various construction scenarios and provide data support for subsequent research.

4.2. Building the CMA dataset

According to the definition of unsafe actions, videos about construction sites are collected from YouTube through keywords of “Construction” and “Behavior”. However, only a small portion of videos are related to the defined actions so the following two methods are used to enrich the dataset.

First, additional daily actions which have similar characteristics to the defined behaviors are added to the CMA dataset. This idea is inspired by contrastive learning [79,80], which aims to make similar objects close in the space of feature embeddings. For example, video clips of “wear motorcycle helmet” are added for the “Wear Helmet” action because “wearing a motorcycle helmet” has the most similarity to “Wear Helmet” compared with other daily actions. When video data of such actions on construction sites are sufficient, videos containing similar daily actions (e.g., wearing) and elements (e.g., hard hat) can help the deep learning model to learn similar feature patterns. In addition, including video data in non-construction scenes introduces variances into data distribution, which can increase the robustness of the model when the data change.

Second, several data augmentation methods, such as horizontal flipping, rotation, and random pixel masking are implemented to further enrich the data. The methods are widely recognized tools to increase the number of training samples while improving the generalization ability of models [38,81]. Some examples of video clips after data augmentation are shown in Fig. 3. For a raw clip $clip^i, i \in 1 \dots N$, clips generate using the augmentation methods are grouped as $G^i = (clip^i, clip_{flip}^i, clip_{rotate}^i, clip_{mask}^i, \dots)$. Finally, the dataset $G = (G^1, \dots, G^i, \dots, G^N)$ can be established, where N is the number of

Table 1

Definition of worker behavior and the number of clips for each behavior in our CMA dataset.

Behavior	Description	Number of clips
Climb Ladder	Climb from the bottom of the ladder to the top of the ladder	140
Fall Down	Change from a standing position to a lying down position	315
Smoke	Smoke cigarettes in working area	365
Stand	Stay in the working area and do nothing for a period of time	315
Talk	Two or more workers communicate in dangerous place	160
Walk	Move slowly in the working area	195
Wear Helmet	The process of the worker putting the helmet on the head	105

Table 2

CMA dataset statistics.

Video clips	Class	Duration	Resolution	Frame rate
1595	7	1s-9s	400 × 300 - 1920 × 1080	30

groups of raw clips. By referring to the method proposed in UCF101 dataset [82], the N groups are divided into two splits, one for training with two-thirds of the full data, and the other for validation and testing with one-third of the full data. This can ensure data in each group not being confused with data in other groups.

4.3. Statistic

The statistics of the CMA dataset are summarized in Table 2. The dataset defines seven classes of behaviors of workers on construction sites. For each class, there are 105–365 clips, and the total number of data samples is 1595. The duration of each clip is 1–9 s with a resolution between 400 × 300 - 1920 × 1080. Each video clip contains only one action and has a fixed frame rate of 30 frames per second. The specific data proportion of each category is shown in Fig. 4. In the dataset, action “Smoke” accounts for the largest proportion (22.9%), whereas the action “Wear Helmet” accounts for the smallest proportion (6.6%). The bar chart in Fig. 5 demonstrates the number of clips in each action. The colors of each bar illustrate durations of different clips belonging to that class. Most of the clips last 2–4 s.

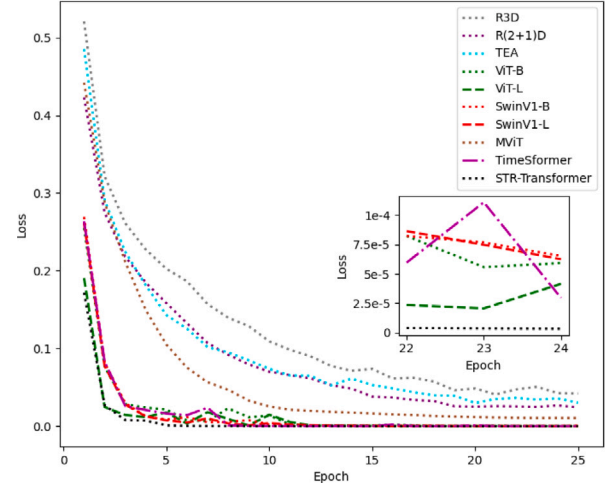
5. Experiments

In this section, the STR-Transformer was compared with other SOTA methods on the standard CMA dataset. Then, ablation experiments were conducted to further verify the effectiveness of the model in detail. Experiments were performed on a single NVIDIA GeForce RTX 3090 GPU. Code is based on [83–86] and will be published at <https://github.com/S1mpleyang/ConstructionActionRecognition>.

5.1. Implementation details

Training. Different video clips had varying durations (30–270 frames), sending all frames into the model would result in a large number of computation operations and long computing time. To address the issue, according to the TSN method [20], a fixed number of frames were filtered from video clips to represent the whole video, thereby reducing computation costs without losing too much information. Specifically, the input video was divided into k segments of equal durations and then one frame was randomly sampled from each segment to get a sequence with k frames. In the experiments, k was set to be 8 and 16 following STM [46]. The frames were resized to 256 × 256 and randomly cropped to 224 × 224 for model training. In other words, the input size of the model was $B \times k \times 3 \times 224 \times 224$, where B is the batch size and k is the number of frames, (3, 224, 224) indicate the number of channels, height, width of the input frame, respectively.

The SwimTransformer pretrained on Imagenet-21k [87] and R(2+1)D pretrained on Kinetics [88] were adopted as the SFE and TFE in the STR-Transformer, respectively. As for the RPFFM, the number

**Fig. 6.** Losses of SOTA methods during training.

of LSTM layer was set to be 1. The dropout ratio was set to be 0.5 to avoid overfitting. The AdamW optimizer was applied [89] with a base learning rate of 1×10^{-5} for pre-trained parameters. The online learning strategy was used, where the batch size was 1. The number of training epochs was 25 for all models. The learning rate declined with a factor of 10 at 10 and 20 epochs. The BCEWithLogitsLoss was used to compute the distance between the current and expected output of the model, which is defined as.

$$Loss(x_i, y_i) = -w_i[y_i \times \log \sigma(x_i) + (1 - y_i) \times \log(1 - \sigma(x_i))] \quad (21)$$

where x_i is the predicted value; y_i is the label value; σ represents the sigmoid function; w_i is the weight vector of i th input, which was set to be 1.

Testing. The spatial dimension of clips was resized to 256 × 256 pixels and cropped to 224 × 224. For the temporal domain, the full-length video was sampled 10 times to compute the softmax scores individually, which are standard procedures in the field of computer vision [20,43,45]. The averaged softmax scores of all clips were taken for the final prediction.

5.2. Results

In this section, using the standard benchmarking dataset CMA, the performance of the STR-Transformer was evaluated and compared with three types of mainstream methods, namely, (1) CNN-based methods: TSN, Temporal Shift Module (TSM), SlowFast, 3D Residual ConvNet (R3D), R(2+1)D and Temporal Excitation and Aggregation (TEA); (2) Transformer-based methods: Vision Transformer Base/Large (ViT-B/L), Swin Transformer Base/Large (Swin-B/L), Video Vision Transformer Large (ViViT-L), Multiview Transformers Base (MTViT-B), Multiscale Vision Transformers Base (MViT-B) and TimeSformer; (3) LSTM-based models: CNN+LSTM reproduced according to Ding et al. [25], where the number of LSTM was set to be 2 and 5. In addition, for SlowFast,

Table 3

Comparison results of per-class precision and average precision of STR-Transformer with other SOTA methods on the CMA dataset.

Method	NOF	Climb Ladder	Fall Down	Smoke	Stand	Talk	Walk	Wear Helmet	AP
TSN [20]	8	86.6%	70.0%	96.0%	88.0%	46.7%	76.3%	68.8%	76.1%
TSM [48]	8	75.5%	88.0%	99.6%	83.2%	60.0%	92.7%	82.2%	83.0%
SlowFast [43]	4+16	82.2%	81.0%	99.7%	88.1%	42.2%	99.8%	53.3%	78.1%
R3D [40]	8	75.1%	76.3%	92.1%	87.3%	34.2%	73.2%	55.3%	70.5%
	16	85.5%	81.4%	96.4%	95.6%	55.6%	87.9%	66.7%	81.3%
R(2+1)D [42]	8	88.2%	72.2%	98.4%	90.8%	63.1%	87.2%	73.7%	81.9%
	16	73.3%	82.0%	93.3%	96.0%	62.2%	92.7%	75.6%	82.1%
TEA [45]	8	84.4%	75.0%	94.0%	81.0%	66.7%	85.4%	77.8%	80.6%
	16	81.8%	52.8%	84.5%	90.6%	62.2%	67.5%	76.0%	73.6%
ViT-B [55]	8	70.2%	81.2%	90.6%	91.0%	60.0%	87.6%	67.5%	78.3%
	16	68.9%	67.6%	95.4%	87.0%	52.4%	90.1%	75.5%	76.7%
ViT-L [55]	8	60.1%	97.0%	96.8%	91.2%	66.7%	83.6%	77.8%	81.9%
	16	73.8%	91.2%	97.0%	93.7%	66.7%	81.8%	82.2%	83.7%
Swin-B [71]	8	77.1%	91.1%	96.8%	92.2%	66.3%	87.0%	82.2%	84.7%
	16	77.4%	87.0%	99.1%	89.2%	67.4%	83.6%	78.5%	83.1%
Swin-L [71]	8	88.9%	93.5%	97.3%	94.2%	66.7%	93.6%	77.8%	87.4%
	16	88.3%	89.0%	94.3%	88.0%	66.7%	99.7%	77.8%	86.3%
MViT-B [63]	8	76.7%	83.0%	94.6%	89.0%	60.0%	94.5%	73.3%	81.6%
	16	73.3%	90.0%	99.3%	99.0%	60.0%	98.1%	75.5%	85.0%
TimeSformer [61]	8	86.6%	77.0%	94.0%	96.0%	62.2%	87.3%	77.8%	82.9%
	16	77.8%	70.0%	89.8%	94.0%	64.4%	80.0%	72.8%	78.4%
STR-Transformer	8	91.1%	91.2%	99.5%	95.0%	66.7%	96.8%	82.2%	88.7%
	16	88.9%	93.8%	98.5%	94.0%	66.7%	86.5%	77.8%	86.6%

Table 4

Comparison results of STR-Transformer and other SOTA methods in FLOPs, Params and inference time.

Method	Pretrain	AP	FLOPs	Params	Time
$CNN + LSTM_2$ [25]	IN-1K	36.4%	22.8	22.6	0.055
$CNN + LSTM_5$	IN-1K	N/A	22.8	22.6	0.104
TSN	K400	76.1%	33.0	24.3	N/A
TSM	K400	83.0%	32.9	24.3	N/A
SlowFast	K400	78.1%	36.6	34.4	N/A
R3D	K400	70.5%	109.8	47.0	0.008
R(2+1)D	K400	81.9%	117.2	47.0	0.015
TEA	K400	80.6%	35.0	24.9	0.028
ViT-B	IN-21K	78.3%	134.8	85.9	0.008
ViT-L	IN-21K	81.9%	477.2	303.4	0.039
Swin-B	IN-21K	84.7%	121.0	87.0	0.020
Swin-L	IN-21K	87.4%	272.1	195.4	0.032
ViViT-L [60]	–	N/A	995.0	N/A	N/A
MTV-B [64]	–	N/A	199.5	310.0	N/A
MViT-B	K400	81.6%	70.5	36.6	0.021
TimeSformer	IN-21K	82.9%	178.6	113.9	0.036
STR-Transformer	IN-21K	88.7%	202.3	121.4	0.061

TSN, and TSM, the training strategy in 202 [85] was applied due to their pre-defined training requirements, whereas other SOTA methods adopted the same training strategy introduced in this study.

The loss curves of SOTA methods during training are shown in Fig. 6. The losses of Transformer-based models drop rapidly and converge at around 10 epochs except MViT-B. The training loss of the STR-Transformer at 25 epoch is $3e-6$, which is the smallest among all models. Other SOTA methods (e.g., R3D, R(2+1)D, TEA) converge slowly, and the final losses remain around $2e-2$. These methods can still reach good prediction performance after a longer training period, at the cost of excessive computation resources and time. Thus, the STR-Transformer outperforms the others in terms of the convergence rate and demands of resources.

The performance comparison between the STR-Transformer with other SOTA methods is listed in Table 3. NOF represented the number of frames in the input clip and the average precision (AP) was used as the metric of prediction performance. Other important metrics, namely, floating point operations (FLOPs), number of parameters (Params), and inference time (Time) are summarized in Table 4. These methods were pretrained on Kinetics-400 (K400) [88] or ImageNet-1K/ImageNet-21K (IN-1K/IN-21K) [87,90] datasets. Calculations of Params and FLOPs

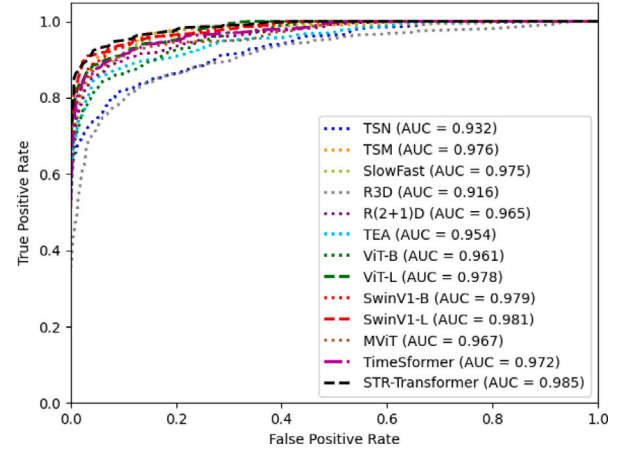


Fig. 7. Receiver Operating Characteristic (ROC) curves of individual models and their area under the curve (AUC). A larger AUC means a better performance of the model.

were based on the pytorch-Opcounter [91], and the shape of input clip was set to be $1 \times 8 \times 3 \times 224 \times 224$. Magnitudes are Giga(10^9) for FLOPs and Mega(10^6) for Params. The inference Time is defined as the time (seconds) the model takes to produce the prediction of each video clip. N/A meant no data was acquired.

As shown in Table 3, it can be argued that the STR-Transformer can achieve 88.7% and 86.6% average precision with 8- and 16-frame inputs, outperforming other SOTA methods. When using fewer input frames (8 frames), the effect is improved by 2.1% compared to using more input frames (16 frames). In general, increasing the length of the input frame can provide more information to the model. For some models, e.g., R3D, and MViT-B, such strategy improves the average precision, which, however, is not the case for other models, e.g., TEA, and TimeSformer, where using 8-frame inputs results in better performance. The STR-Transformer also performs better with 8-frame inputs, to-some-extent indicating the superior ability in terms of extracting temporal and spatial features even the number of input frames is limited.

As shown in Table 4, it can be argued that the Transformer-based models have the largest FLOPs and Params compared with the CNN-based and LSTM-based models. Among these Transformer-based models, ViViT-L has the largest FLOPs (995.0 GFLOPs), almost 5 times

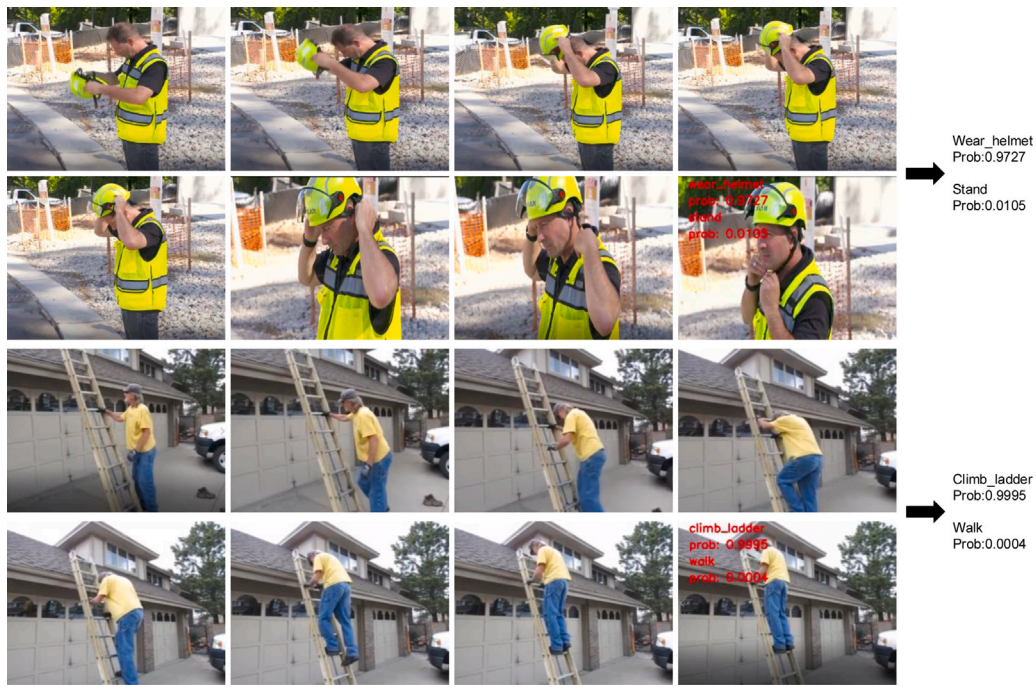


Fig. 8. Demo videos produced by STR-Transformer. Top left in the final frame shows the classification results and probability scores for the top-2 actions.

that of the STR-Transformer. MTV-B has the largest Params (310.0 MB), about 2.5 times that of the STR-Transformer. Compared with Swin-B, the baseline method, the precision of the STR-Transformer is 4.0% higher using 8-frame inputs, and the FLOPs and Params increase from 121.0 to 202.3 GFLOPs and from 87.0 to 121.4 MB, respectively. Compared with Swin-L, which reaches the best performance among the SOTA models, the STR-Transformer still gains a precision improvement of 1.3%, and the FLOPs and Params also decrease from 272.1 to 202.3 GFLOPs and from 195.4 to 121.4 MB, respectively. In other words, compared with these transformer-based models, the STR-Transformer can improve the performance of action recognition while increasing model complexity by an acceptable extent.

As shown in Table 4, it can be argued that the transformer-based method requires slightly more training time than the CNN-based method, but the LSTM-based methods are the most time-consuming ones. The inference Time of the STR-Transformer is 0.061 s per clip. Although it is larger than that of MViT-B (0.021 s) and TimeSformer (0.036 s), it is still acceptable for most practical applications. As for the LSTM-based model (CNN+LSTM in Ding et al. 25), the inference Time is 0.055 when the number of LSTM was set to be 2, which is already close to the STR-Transformer. As the depth of LSTM-based methods increases, the time needed will further increase. Thus, the STR-Transformer is more suitable for actual implementation in construction projects than LSTM-based methods.

Additionally, it is worth noting that in the inputs, the STR-Transformer converts 8 RGB frames into 8 RGB difference frames before feeding them to SFE and TFE for feature extraction. As such, when evaluating FLOPs and inference time, the STR-Transformer model used 16 frames, while other models used 8 RGB frames. Thus, the FLOPs and inference time of the other models should be doubled, and the complexity of the STR-Transformer can be regarded as low.

Moreover, the Receiver Operating Characteristics (ROC) curves of models in the experiments were drawn in Fig. 7, while the area under the curve (AUC) were evaluated for comprehensive comparison. It is obvious that the STR-Transformer has the highest AUC among all models.

The demo video produced by the STR-Transformer was demonstrated in Fig. 8. The two classification results with the highest prediction scores were presented at the top left in the last frame.

All models perform well when classifying some actions, such as “Smoke” and “Stand”. However, they also all fail to recognize some actions, such as “Talk”, where the precision is less than 67.4%. The SlowFast performs best in terms of recognizing “Smoke” and “Walk”, the MViT-B performs best in terms of recognizing “Stand”, and the Swin-B performs best when identifying “Talk”. To better investigate the performance of different models classifying different actions, two confusion matrices were generated for the baseline method and STR-Transformer, as shown in Figs. 9 and 10, respectively. As mentioned, for both models, the “Talk” action is the most challenging one to classify, and the “Smoke” is the easiest one. Furthermore, both models can occasionally confuse “Talk” and “Wear Helmet” with “Stand”, but hardly recognize “Stand” as “Talk” and “Wear Helmet”. As “Stand” is usually the premise of “Talk” or “Wear Helmet”, this phenomena can shed light on how deep learning models learn features in similar classes. It is supposed that the problem can be solved by dividing the action into more detailed classes or by improving the feature extraction capacity of the models.

5.3. Ablation study

Ablation experiments were conducted to demonstrate the effectiveness different parts of the model. The models were trained and tested with 8-frame inputs using samples in the CMA dataset, and the training strategy was same as the STR-Transformer. The results are shown in Table 5. Feature Fusion represented the method to fuse spatial and temporal features. ADD and RPFPM represented two operations to fuse two kinds of features. Pretrain represented the pre-trained weights used by SFE+ TFE. Data Type represented the type of input video clip of SFE+ TFE.

Effective of spatial feature encoder One CNN-based model (i.e., Resnet-50 [92]) and two Transformer-based models (i.e., ViT-B and Swin-B) were selected as the SFE. As summarized in the first part of Table 5, Transformer-based models perform better than CNN-based models due to the unique attention mechanism (74.7%, 86.2% vs. 49.7%). The Swin-B applied a sliding window strategy, which can restrict the self-attention computations to non-overlapping local windows, while allowing cross-window connections. This results in higher precision (86.2% vs. 74.7%) and efficiency (202.2 vs. 216.1 GFLOPs) compared to ViT-B.

Table 5
Ablation experiments results of STR-Transformer on CMA dataset.

SFE	TFE	Feature fusion	Pretrain	Data type	AP	FLOPs	Params	Time
Resnet-50 [92]	R(2+1)D-18	ADD	IN-1K+K400	RGB+RGB DIFF	49.7%	114.2	56.0	0.014
ViT-B	R(2+1)D-18	ADD	IN-21K+K400	RGB+RGB DIFF	74.7%	216.1	117.6	0.023
Swin-B	R(2+1)D-18	ADD	IN-21K+K400	RGB+RGB DIFF	86.2%	202.2	118.7	0.038
Swin-B	None	None	IN-21K	RGB+RGB DIFF	84.7%	121.0	87.0	0.020
Swin-B	Resnet-18	ADD	IN-21K+IN-1K	RGB+RGB DIFF	85.4%	135.6	99.1	0.026
Swin-B	Densenet-121 [93]	ADD	IN-21K+IN-1K	RGB+RGB DIFF	85.7%	143.9	95.4	0.042
Swin-B	R3D-18	ADD	IN-21K+K400	RGB+RGB DIFF	85.5%	206.3	120.6	0.031
Swin-B	R(2+1)D-18	ADD	IN-21K+K400	RGB+RGB DIFF	86.2%	202.2	118.7	0.038
Resnet-50	R(2+1)D-18	RPFFM	IN-1K+K400	RGB+RGB DIFF	61.9%	114.2	61.6	0.063
ViT-B	R(2+1)D-18	RPFFM	IN-21K+K400	RGB+RGB DIFF	83.1%	216.1	119.6	0.051
Swin-B	R(2+1)D-18	RPFFM	IN-21K+K400	RGB+RGB DIFF	88.7%	202.3	121.4	0.061
Resnet-50	R(2+1)D-18	RPFFM	IN-1K+K400	RGB+RGB	56.3%	114.2	61.6	0.063
ViT-B	R(2+1)D-18	RPFFM	IN-21K+K400	RGB+RGB	80.6%	216.1	119.6	0.051
Swin-B	R(2+1)D-18	RPFFM	IN-21K+K400	RGB+RGB	85.9%	202.3	121.4	0.061
Swin-B	R(2+1)D-18	RPFFM	IN-21K+K400	RGB+Optical Flow	N/A	202.3+N/A	121.4+5.3	0.061+0.084

Table 6
Ablation experiments results of FLOPs, Params and inference Time of RPFFM.

Method	CAM	RAM	DTM	AP	FLOPs	Params	Time
SFE+TFE	–	–	–	86.2%	202.2	118.7	0.038
SFE+TFE	✓	–	–	87.1%	202.2	119.4	0.038
SFE+TFE	✓	✓	–	87.5%	202.3	119.7	0.039
SFE+TFE	✓	✓	✓	88.7%	202.3	121.4	0.061

Effective of temporal feature encoder This experiment aims to verify whether the TFE can improve model performance, while finding out the model suitable as the TFE. Two 2D-CNN-based models (i.e., Resnet-18 and Densenet-121 [93]), two 3D-CNN-based models (i.e., R(2+1)D-18 and R3D-18) were selected as the TFE. Features of SFE and TFE were fused by an additional ADD operation. The Results are shown in the second part of Table 5. Models equipped with TFE can achieve improvements of different degrees compared to the baseline model (i.e., 85.4%, 85.7%, 85.5%, 86.2% vs. 84.7%). Moreover, the performance of using a 3D-CNN-based model as TFE is not significantly different from the case using a 2D-CNN-based model (85.5%, 86.2% vs. 85.4%, 85.7%). As such, the R(2+1)D-18 was selected as TFE in the STR-Transformer because it can reach the highest AP of 86.2%.

Effective of multimodal data The effect of using multi-modal data on performance was investigated. Three combinations of modals were tested, i.e., RGB+RGB, RGB+RGB DIFF, and RGB+Optical Flow. The method in Teed and Deng [94] was applied to extract Optical Flow from videos. As shown in the third and fourth parts of Table 5, models using multimodal data (RGB frames and RGB difference frames) perform better than the models using only RGB frames (83.1%, 88.7% vs. 80.6%, 85.9%). This is because the spatial and temporal features carried by RGB difference frames are different from those in RGB frames, which facilitate feature extraction and learning. However, collecting and processing multi-modal data requires more resources. For example, the time to extract RGB difference frames is negligible, while extracting optical flow requires additional 0.084s per clip, which is even larger than that of the model making a prediction. Thus, multi-modal data should be used with caution, taking hardware conditions in specific projects into account.

Effective of relative position feature fusion model The RPFFM consists of three submodules, i.e., CAM, RAM, and DTM. They were employed to fuse spatial and temporal features from SFE and TFE. The results of the ablation experiment are listed in Table 6. One can conclude that the CAM can add channel attention weights to features so that representative features are more prominent, thereby improving AP of the baseline model from 86.2% to 87.1%. Additionally, after implementing the RAM, the correlation of each part of a feature is enhanced, and the AP is again increased from 87.1% to 87.5%. Finally, the DTM can also enhance the spatial-temporal representation

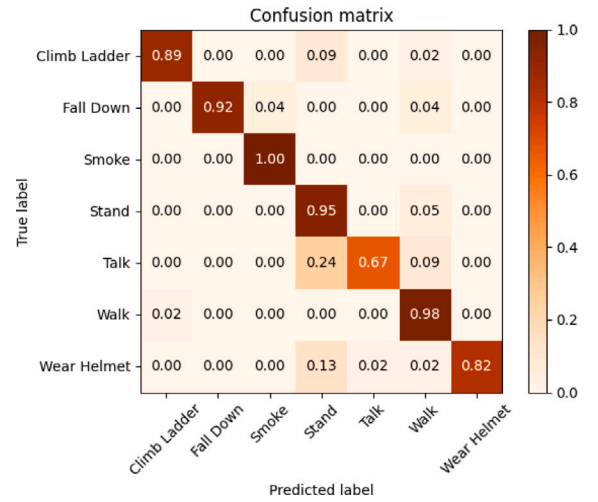


Fig. 9. Confusion matrix generated by our STR-Transformer.

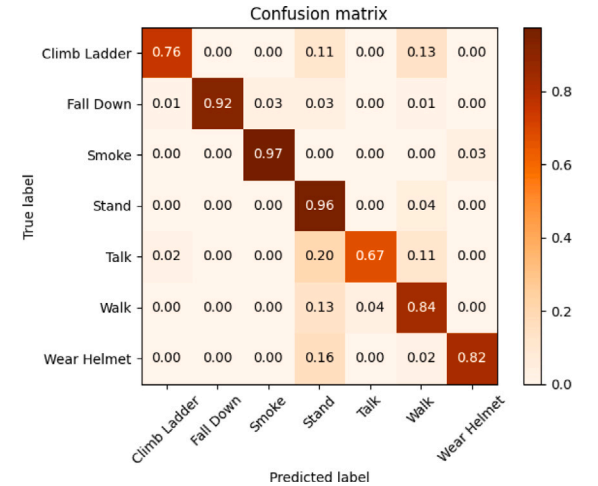


Fig. 10. Confusion matrix generated by baseline model.

of features, gaining an improvement of a 1.2% AP. Meanwhile, as presented in Table 6, it can be argued that the FLOPs and Params of RPFFM are 0.1 GFLOPs and 2.7 MB respectively, which is very small compared to the whole model. However, the RPFFM increases inference Time by 0.023 s, because the DTM is relatively time-consuming. The

experiments prove that the RPFFM can effectively fuse temporal and spatial features and improve the performance of the STR-Transformer

6. Discussion

The contribution of this study is three folds. First, recent advances in CV techniques in the field of action recognition and their applications in monitoring unsafe behaviors on construction sites are discussed. Several limitations of existing studies are identified: (1) Signal-based methods usually ignore important visual information. (2) Image-based methods often focus on detecting and locating PPE and construction equipment in images while ignoring consecutive actions that require temporal feature extraction for accurate identification. (3) Conventional two-stage models and LSTM-based models are time-consuming. (4) Common CNN-based models cannot model long-range spatial-temporal features. (5) Transformer is difficult to train from scratch on small datasets. (6) Some two-stream models focus on extracting spatial and temporal features separately while ignoring fusing the two types of features.

Second, to address these limitations, a Transformer-based model named STR-Transformer is proposed in this study. This model can extract adequate spatial and temporal features from videos and consists of three parts: (1) a SFE based on the Transformer model to extract features from frames, where the unique self-attention mechanism aggregates information from the whole frame. The original architecture of Transformer remains unchanged to facilitate transferring weights pre-trained in other large datasets to the STR-Transformer and fine-tuning on small datasets. Experiments demonstrate that the STR-Transformer achieves an improvement of 8.6% average precision compared with TSN, and has the highest convergence rate. (2) a TFE based on a 3D CNN to effectively model and enhance temporal features from consecutive video frames. This strategy improves the average precision from 84.7% to 86.2%. (3) a RPFFM to fuse and enhance the spatial and temporal features. With the RPFFM, the average precision of the whole model gains a further improvement of 2.5%. Experiments demonstrate that the STR-Transformer outperforms the SOTA methods.

When applied in the construction industry, the STR-Transformer has the following practical contributions: (1) Compared with traditional manual monitoring, this model identifies unsafe behaviors of construction workers automatically from surveillance videos. As such, the challenges of subjectivity and limited coverage faced by manual surveillance can be addressed with reduced management costs. (2) Compared to recent methods that identify unsafe worker behaviors automatically using visual data, the STR-Transformer incorporates both spatial and temporal features, therefore achieving higher performance and reliability. The model can create value for project teams by detecting risky behaviors that could be omitted by previous image- or video-based methods in the industry. (3) In addition, the STR-Transformer keeps the initial structure of Transformer. Thus, the model pre-trained on a large dataset can be transferred to the task of unsafe behavior detection. Experiments show that the STR-Transformer converges quickly on small datasets, indicating that it could also be fine-tuned and generalized for other types of unsafe behaviors and possibly other safety management tasks. For example, hazardous objects and risk areas could be identified with a small number of labeled samples and a short training period. This enables flexible and fast deployment of the model to meet practical demands of a project.

In addition, a construction worker behavior recognition dataset named CMA is proposed. Creating a dataset for general objectives is expensive and this has been a chronic problem in the construction community [27]. Currently, different studies use different datasets to train models for unsafe action identification, which leads to limited cross-comparisons and thus low confidence in existing models. To address this issue, the CMA includes seven classes of actions and 1595 video clips. Each clip in this dataset depicts only one action, making it convenient for other models to train and test on it without modification.

In a word, this dataset has great potential to serve other video-based studies on construction sites.

This work also has several limitations and could be improved in the future. First, with the improvement of its average accuracy, the complexity of the STR-Transformer increases because Transformer has higher complexity than CNN. Subsequently work will be done to make this model lighter and more efficient. Second, the definition of worker actions in this paper is not specific enough. More detailed worker behaviors will be included by expanding the dataset and divided into detailed behavioral categories for more precise detection and classification. At last, due to the lack of video data, some similar actions in daily life are added to the CMA dataset. It should be admitted that even similar actions have slight differences so more related video data of construction scenes will be collected into the CMA dataset in future work.

7. Conclusions

Construction remains one of the most dangerous industries in the world. Due to the complex environment of construction sites and the inefficiency of manual supervision, unsafe behaviors of workers are the main cause of construction accidents. To improve work safety, it is critical to implement computer vision based methods to automatically and effectively detect workers' unsafe behaviors. Currently, such methods are restricted by their poor performance in terms of simultaneously capturing and fusing spatial and long-range temporal features in consecutive video clips, which to a large extent hinders their practical application. Moreover, although AI-based methods for risky behavior detection are continuously proposed recently, the construction industry lacks an especially designed benchmarking dataset. As a result, the effectiveness of these methods cannot be objectively verified, which also hinders their development and implementation.

In this paper, a new model named STR-Transformer is developed for unsafe action recognition of construction workers. First, two SOTA computer vision methods, Transformer and 3D CNN, are used to extract the spatial and temporal features of workers' behaviors in consecutive video clips. Second, a RPFFM is designed to fuse and enhance the two features to further improve the prediction power. Third, a dataset named CMA, which contains 1595 video clips of worker behaviors divided into seven classes, is established to verify the STR-Transformer. The experimental results demonstrate that the proposed model could successfully identify unsafe actions with an average precision of 88.7%, which is 4% better than that of the baseline model. The high precision indicates that the proposed model can be effectively used to identify unsafe actions in videos of construction sites. The results also demonstrate that using multimodal data (RGB and RGB difference frames) to identify unsafe behavior can achieve better results than unimodal data (RGB frames) with an improvement of 2.8%. This is because multimodal data allows the model to extract more types of features therefore improving the prediction. Finally, fusing features using the RPFFM also increases the performance to 88.7%, 2.5% higher than the baseline ADD operation. In addition, the CMA dataset can not only provide valuable benchmarking data for subsequent studies related to action recognition of construction workers, but also can be widely used in other research fields such as worker tracking and motion understanding.

There are several further research directions. First, the current STR-Transformer can reach high precision of behavior recognition, but its complex structure can restrict its application due to limited computation resources in practical projects. Thus, future work will focus on balancing model capacity and computation costs, while developing lightweight models to meet the practical demands of real-time response and safety warning on sites. Second, given the value of using multimodal data, it is interesting and meaningful to investigate whether other multimodal data, such as optical flow frames, depth frames, and multi-camera videos, will also improve the model performance.

Third, as the feature fusion module produces direct inputs to the final predictor, it is necessary to investigate and design more effective fusing methods to take full use of features in multimodal data. Finally, in this study, daily actions are added to the CMA dataset due to the lack of videos on construction sites. Therefore, more videos of unsafe behavior on construction sites will be collected, pre-processed, and added to the dataset in the future. To this end, it can be concluded the proposed STR-Transformer is an effective tool for on-site safety monitoring, which can release supervisors from tedious manual supervision tasks, reduce accident rates, and save safety management costs in construction projects.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

This article provides a GitHub link to the code and data.

Acknowledgments

The authors would like to acknowledge Wenjing Zhang and Zhenghan Liao for their help. This paper is financially supported by the Nature and Science Foundation of China under grants 52077213, 62003332, and 52208324, the Shenzhen Science and Technology Innovation Commission under grants GJHZ20210705141401005 and Shaoguan “Nanling Talent Group” Grant.

References

- [1] Briefing on the national production safety situation in the construction industry, 2022, https://www.mem.gov.cn/gk/tzgg/tb/201807/t20180725_230568.shtml (Accessed 2, June, 2022).
- [2] National Census of Fatal Occupational Injuries in 2020, U.S. Bureau of Labor, Statistics, Washington, DC, USA, 2022, <https://www.bls.gov/news.release/pdf/foi.pdf> (Accessed 2, June, 2022).
- [3] J. Liu, H. Luo, H. Liu, Deep learning-based data analytics for safety in construction, *Autom. Constr.* 140 (2022) 104302, <http://dx.doi.org/10.1016/j.autcon.2022.104302>.
- [4] S. Wu, L. Hou, G.K. Zhang, H. Chen, Real-time mixed reality-based visual warning for construction workforce safety, *Autom. Constr.* 139 (2022) 104252, <http://dx.doi.org/10.1016/j.autcon.2022.104252>.
- [5] Z. Jiang, D. Fang, M. Zhang, Understanding the causation of construction workers' unsafe behaviors based on system dynamics modeling, *J. Manage. Eng.* 31 (6) (2015) 04014099, [http://dx.doi.org/10.1061/\(ASCE\)JME.1943-5479.0000350](http://dx.doi.org/10.1061/(ASCE)JME.1943-5479.0000350).
- [6] H.W. Heinrich, et al., *Industrial Accident Prevention. A Scientific Approach*, New York & London: McGraw-Hill Book Company, Inc., ISBN: 978-0070280588, 1941.
- [7] S. Taneja, B. Akinci, J.H. Garrett, L. Soibelman, E. Ergen, A. Pradhan, P. Tang, M. Berges, G. Atasoy, X. Liu, et al., Sensing and field data capture for construction and facility operations, *J. Constr. Eng. Manag.* 137 (10) (2011) 870–881, [http://dx.doi.org/10.1061/\(ASCE\)CO.1943-7862.0000332](http://dx.doi.org/10.1061/(ASCE)CO.1943-7862.0000332).
- [8] Z. Li, D. Li, Action recognition of construction workers under occlusion, *J. Build. Eng.* 45 (2022) 103352, <http://dx.doi.org/10.1016/j.jobte.2021.103352>.
- [9] I. Awolusi, E. Marks, M. Halliwell, Wearable technology for personalized construction safety monitoring and trending: Review of applicable devices, *Autom. Constr.* 85 (2018) 96–106, <http://dx.doi.org/10.1016/j.autcon.2017.10.010>.
- [10] M. Jung, S. Chi, Human activity classification based on sound recognition and residual convolutional neural network, *Autom. Constr.* 114 (2020) 103177, <http://dx.doi.org/10.1016/j.autcon.2020.103177>.
- [11] S.S. Bangaru, C. Wang, S.A. Busam, F. Aghazadeh, ANN-based automated scaffold builder activity recognition through wearable EMG and IMU sensors, *Autom. Constr.* 126 (2021) 103653, <http://dx.doi.org/10.1016/j.autcon.2021.103653>.
- [12] K. Lee, S. Han, Convolutional neural network modeling strategy for fall-related motion recognition using acceleration features of a scaffolding structure, *Autom. Constr.* 130 (2021) 103857, <http://dx.doi.org/10.1016/j.autcon.2021.103857>.
- [13] S. Paneru, I. Jeelani, Computer vision applications in construction: Current state, opportunities & challenges, *Autom. Constr.* 132 (2021) 103940, <http://dx.doi.org/10.1016/j.autcon.2021.103940>.
- [14] N.D. Nath, A.H. Behzadan, S.G. Paal, Deep learning for site safety: Real-time detection of personal protective equipment, *Autom. Constr.* 112 (2020) 103085, <http://dx.doi.org/10.1016/j.autcon.2020.103085>.
- [15] R. Xiong, P. Tang, Pose guided anchoring for detecting proper use of personal protective equipment, *Autom. Constr.* 130 (2021) 103828, <http://dx.doi.org/10.1016/j.autcon.2021.103828>.
- [16] M. Yang, Z. Yang, Y. Guo, S. Su, Z. Fan, A novel YOLO based safety helmet detection in intelligent construction platform, in: *Intelligent Equipment, Robots, and Vehicles*, Springer, 2021, pp. 268–275, http://dx.doi.org/10.1007/978-981-16-7213-2_26.
- [17] E. Chian, W. Fang, Y.M. Goh, J. Tian, Computer vision approaches for detecting missing barricades, *Autom. Constr.* 131 (2021) 103862, <http://dx.doi.org/10.1016/j.autcon.2021.103862>.
- [18] W. Fang, B. Zhong, N. Zhao, P.E. Love, H. Luo, J. Xue, S. Xu, A deep learning-based approach for mitigating falls from height with computer vision: Convolutional neural network, *Adv. Eng. Inform.* 39 (2019) 170–177, <http://dx.doi.org/10.1016/j.aei.2018.12.005>.
- [19] D. Tran, L. Bourdev, R. Fergus, L. Torresani, M. Paluri, Learning spatiotemporal features with 3D convolutional networks, in: *Proceedings of the IEEE International Conference on Computer Vision*, 2015, pp. 4489–4497, <http://dx.doi.org/10.1109/ICCV.2015.510>.
- [20] L. Wang, Y. Xiong, Z. Wang, Y. Qiao, D. Lin, X. Tang, L.V. Gool, Temporal segment networks: Towards good practices for deep action recognition, in: *Proceedings of the European Conference on Computer Vision*, Springer, 2016, pp. 20–36, http://dx.doi.org/10.1007/978-3-319-46484-8_2.
- [21] D. Roberts, W. Torres Calderon, S. Tang, M. Golparvar-Fard, Vision-based construction worker activity analysis informed by body posture, *J. Comput. Civ. Eng.* 34 (4) (2020) 04020017, [http://dx.doi.org/10.1061/\(ASCE\)CP.1943-5487.0000898](http://dx.doi.org/10.1061/(ASCE)CP.1943-5487.0000898).
- [22] C. Ding, S. Wen, W. Ding, K. Liu, E. Belyaev, Temporal segment graph convolutional networks for skeleton-based action recognition, *Eng. Appl. Artif. Intell.* 110 (2022) 104675, <http://dx.doi.org/10.1016/j.engappai.2022.104675>.
- [23] S. Subedi, N. Pradhananga, Sensor-based computational approach to preventing back injuries in construction workers, *Autom. Constr.* 131 (2021) 103920, <http://dx.doi.org/10.1016/j.autcon.2021.103920>.
- [24] X. Luo, H. Li, Y. Yu, C. Zhou, D. Cao, Combining deep features and activity context to improve recognition of activities of workers in groups, *Comput.-Aided Civ. Infrastruct. Eng.* 35 (9) (2020) 965–978, <http://dx.doi.org/10.1111/mice.12538>.
- [25] L. Ding, W. Fang, H. Luo, P.E. Love, B. Zhong, X. Ouyang, A deep hybrid learning model to detect unsafe behavior: Integrating convolution neural networks and long short-term memory, *Autom. Constr.* 86 (2018) 118–124, <http://dx.doi.org/10.1016/j.autcon.2017.11.002>.
- [26] S. Jung, J. Jeoung, H. Kang, T. Hong, 3D convolutional neural network-based one-stage model for real-time action detection in video of construction equipment, *Comput.-Aided Civ. Infrastruct. Eng.* 37 (1) (2022) 126–142, <http://dx.doi.org/10.1111/mice.12695>.
- [27] X. Luo, H. Li, D. Cao, Y. Yu, X. Yang, T. Huang, Towards efficient and objective work sampling: Recognizing workers' activities in site surveillance videos with two-stream convolutional networks, *Autom. Constr.* 94 (2018) 360–370, <http://dx.doi.org/10.1016/j.autcon.2018.07.011>.
- [28] W. Fang, L. Ding, H. Luo, P.E. Love, Falls from heights: A computer vision-based approach for safety harness detection, *Autom. Constr.* 91 (2018) 53–61, <http://dx.doi.org/10.1016/j.autcon.2018.02.018>.
- [29] S. Hochreiter, J. Schmidhuber, Long short-term memory, *Neural Comput.* 9 (8) (1997) 1735–1780, <http://dx.doi.org/10.1162/neco.1997.9.8.1735>.
- [30] M.F. Antwi-Afari, Y. Qarout, R. Herzallah, S. Anwer, W. Umer, Y. Zhang, P. Manu, Deep learning-based networks for automated recognition and classification of awkward working postures in construction using wearable insole sensor data, *Autom. Constr.* 136 (2022) 104181, <http://dx.doi.org/10.1016/j.autcon.2022.104181>.
- [31] Y.-C. Lee, M. Shariatfar, A. Rashidi, H.W. Lee, Evidence-driven sound detection for prenotification and identification of construction safety hazards and accidents, *Autom. Constr.* 113 (2020) 103127, <http://dx.doi.org/10.1016/j.autcon.2020.103127>.
- [32] Q. Fang, H. Li, X. Luo, L. Ding, H. Luo, C. Li, Computer vision aided inspection on falling prevention measures for steeplejacks in an aerial environment, *Autom. Constr.* 93 (2018) 148–164, <http://dx.doi.org/10.1016/j.autcon.2018.05.022>.
- [33] A. Arac, P. Zhao, B.H. Dobkin, S.T. Carmichael, P. Golshani, DeepBehavior: A deep learning toolbox for automated analysis of animal and human behavior imaging data, *Front. Syst. Neurosci.* 13 (2019) 20, <http://dx.doi.org/10.3389/fnsys.2019.00020>.
- [34] F. Zhou, Z. Jiang, Z. Liu, F. Chen, L. Chen, L. Tong, Z. Yang, H. Wang, M. Fei, L. Li, H. Zhou, Structured context enhancement network for mouse pose estimation, *IEEE Trans. Circuits Syst. Video Technol.* 32 (5) (2022) 2787–2801, <http://dx.doi.org/10.1109/TCSVT.2021.3098497>.
- [35] Z. Jiang, F. Zhou, A. Zhao, X. Li, L. Li, D. Tao, X. Li, H. Zhou, Multi-view mouse social behaviour recognition with deep graphic model, *IEEE Trans. Image Process.* 30 (2021) 5490–5504, <http://dx.doi.org/10.1109/TIP.2021.3083079>.

- [36] C. Wu, X. Li, Y. Guo, J. Wang, Z. Ren, M. Wang, Z. Yang, Natural language processing for smart construction: Current status and future directions, *Autom. Constr.* 134 (2022) 104059, <http://dx.doi.org/10.1016/j.autcon.2021.104059>.
- [37] X. Li, C. Wu, F. Xue, Z. Yang, J. Lou, W. Lu, Ontology-based mapping approach for automatic work packaging in modular construction, *Autom. Constr.* 134 (2022) 104083, <http://dx.doi.org/10.1016/j.autcon.2021.104083>.
- [38] K. Simonyan, A. Zisserman, Two-stream convolutional networks for action recognition in videos, *Adv. Neural Inf. Process. Syst.* 27 (2014) <http://dx.doi.org/10.48550/arXiv.1406.2199>.
- [39] C. Feichtenhofer, A. Pinz, A. Zisserman, Convolutional two-stream network fusion for video action recognition, in: *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 2016, pp. 1933–1941, <http://dx.doi.org/10.48550/arXiv.1604.06573>.
- [40] D. Tran, J. Ray, Z. Shou, S.-F. Chang, M. Paluri, Convnet architecture search for spatiotemporal feature learning, 2017, <http://dx.doi.org/10.48550/arXiv.1708.05038>, arXiv preprint [arXiv:1708.05038](http://dx.doi.org/10.48550/arXiv.1708.05038).
- [41] J. Carreira, A. Zisserman, Quo vadis, action recognition? A new model and the kinetics dataset, in: *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 2017, pp. 4724–4733, <http://dx.doi.org/10.1109/CVPR.2017.502>.
- [42] D. Tran, H. Wang, L. Torresani, J. Ray, Y. LeCun, M. Paluri, A closer look at spatiotemporal convolutions for action recognition, in: *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 2018, pp. 6450–6459, <http://dx.doi.org/10.1109/CVPR.2018.00675>.
- [43] C. Feichtenhofer, H. Fan, J. Malik, K. He, Slowfast networks for video recognition, in: *Proceedings of the IEEE International Conference on Computer Vision*, 2019, pp. 6202–6211, <http://dx.doi.org/10.48550/arXiv.1812.03982>.
- [44] C. Feichtenhofer, X3D: Expanding architectures for efficient video recognition, in: *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 2020, pp. 203–213, <http://dx.doi.org/10.48550/arXiv.2004.04730>.
- [45] Y. Li, B. Ji, X. Shi, J. Zhang, B. Kang, L. Wang, Tea: Temporal excitation and aggregation for action recognition, in: *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 2020, pp. 909–918, <http://dx.doi.org/10.48550/arXiv.2004.01398>.
- [46] B. Jiang, M. Wang, W. Gan, W. Wu, J. Yan, Stm: Spatiotemporal and motion encoding for action recognition, in: *Proceedings of the IEEE International Conference on Computer Vision*, 2019, pp. 2000–2009.
- [47] X. Wang, R. Girshick, A. Gupta, K. He, Non-local neural networks, in: *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 2018, pp. 7794–7803, <http://dx.doi.org/10.48550/arXiv.1711.07971>.
- [48] J. Lin, C. Gan, S. Han, TSM: Temporal shift module for efficient video understanding, in: *Proceedings of the IEEE International Conference on Computer Vision*, 2019, pp. 7082–7092, <http://dx.doi.org/10.1109/ICCV.2019.00718>.
- [49] S. Han, S. Lee, A vision-based motion capture and recognition framework for behavior-based safety management, *Autom. Constr.* 35 (2013) 131–141, <http://dx.doi.org/10.1016/j.autcon.2013.05.001>.
- [50] J. Yang, Z. Shi, Z. Wu, Vision-based action recognition of construction workers using dense trajectories, *Adv. Eng. Inform.* 30 (3) (2016) 327–336, <http://dx.doi.org/10.1016/j.aei.2016.04.009>.
- [51] H. Wang, A. Kläser, C. Schmid, C.-L. Liu, Action recognition by dense trajectories, in: *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 2011, pp. 3169–3176, <http://dx.doi.org/10.1109/CVPR.2011.5995407>.
- [52] J. Kim, S. Chi, Action recognition of earthmoving excavators based on sequential pattern analysis of visual features and operation cycles, *Autom. Constr.* 104 (2019) 255–264, <http://dx.doi.org/10.1016/j.autcon.2019.03.025>.
- [53] M. Bügler, A. Borrmann, G. Ogunmakin, P.A. Vela, J. Teizer, Fusion of photogrammetry and video analysis for productivity assessment of earthwork processes, *Comput.-Aided Civ. Infrastruct. Eng.* 32 (2) (2017) 107–123, <http://dx.doi.org/10.1111/mice.12235>.
- [54] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A.N. Gomez, Ł. Kaiser, I. Polosukhin, Attention is all you need, *Adv. Neural Inf. Process. Syst.* 30 (2017) <http://dx.doi.org/10.48550/arXiv.1706.03762>.
- [55] A. Dosovitskiy, L. Beyer, A. Kolesnikov, D. Weissenborn, X. Zhai, T. Unterthiner, M. Dehghani, M. Minderer, G. Heigold, S. Gelly, et al., An image is worth 16x16 words: Transformers for image recognition at scale, 2020, <http://dx.doi.org/10.48550/arXiv.2010.11929>, arXiv preprint [arXiv:2010.11929](http://dx.doi.org/10.48550/arXiv.2010.11929).
- [56] H. Touvron, M. Cord, M. Douze, F. Massa, A. Sablayrolles, H. Jégou, Training data-efficient image transformers & distillation through attention, in: *International Conference on Machine Learning*, PMLR, 2021, pp. 10347–10357, <http://dx.doi.org/10.48550/arXiv.2012.12877>.
- [57] M. Wang, J. Xing, Y. Liu, Actionclip: A new paradigm for video action recognition, 2021, <http://dx.doi.org/10.48550/arXiv.2109.08472>, arXiv preprint [arXiv:2109.08472](http://dx.doi.org/10.48550/arXiv.2109.08472).
- [58] D. Neimark, O. Bar, M. Zohar, D. Asselmann, Video transformer network, in: *Proceedings of the IEEE International Conference on Computer Vision*, 2021, pp. 3163–3172, <http://dx.doi.org/10.1109/ICCVW54120.2021.00355>.
- [59] M. Raghu, T. Unterthiner, S. Kornblith, C. Zhang, A. Dosovitskiy, Do vision transformers see like convolutional neural networks? *Adv. Neural Inf. Process. Syst.* 34 (2021) <http://dx.doi.org/10.48550/arXiv.2108.08810>.
- [60] A. Arnab, M. Dehghani, G. Heigold, C. Sun, M. Lučić, C. Schmid, Vivit: A video vision transformer, in: *Proceedings of the IEEE International Conference on Computer Vision*, 2021, pp. 6836–6846, <http://dx.doi.org/10.48550/arXiv.2103.15691>.
- [61] G. Bertasius, H. Wang, L. Torresani, Is space-time attention all you need for video understanding? 2021, <http://dx.doi.org/10.48550/arXiv.2102.05095>, arXiv preprint [arXiv:2102.05095](http://dx.doi.org/10.48550/arXiv.2102.05095).
- [62] Z. Liu, J. Ning, Y. Cao, Y. Wei, Z. Zhang, S. Lin, H. Hu, Video swin transformer, in: *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 2022, pp. 3202–3211, <http://dx.doi.org/10.48550/arXiv.2106.13230>.
- [63] H. Fan, B. Xiong, K. Mangalam, Y. Li, Z. Yan, J. Malik, C. Feichtenhofer, Multiscale vision transformers, in: *Proceedings of the IEEE International Conference on Computer Vision*, 2021, pp. 6824–6835, <http://dx.doi.org/10.48550/arXiv.2104.11227>.
- [64] S. Yan, X. Xiong, A. Arnab, Z. Lu, M. Zhang, C. Sun, C. Schmid, Multiview transformers for video recognition, in: *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 2022, pp. 3333–3343, <http://dx.doi.org/10.48550/arXiv.2201.04288>.
- [65] I.O. Tolstikhin, N. Houlsby, A. Kolesnikov, L. Beyer, X. Zhai, T. Unterthiner, J. Yung, A. Steiner, D. Keysers, J. Uszkoreit, et al., Mlp-mixer: An all-mlp architecture for vision, *Adv. Neural Inf. Process. Syst.* 34 (2021) 24261–24272, <http://dx.doi.org/10.48550/arXiv.2105.01601>.
- [66] E.A. Shamsabadi, C. Xu, A.S. Rao, T. Nguyen, T. Ngo, D. Dias-da Costa, Vision transformer-based autonomous crack detection on asphalt and concrete surfaces, *Autom. Constr.* 140 (2022) 104316, <http://dx.doi.org/10.1016/j.autcon.2022.104316>.
- [67] W. Wang, C. Su, Automatic concrete crack segmentation model based on transformer, *Autom. Constr.* 139 (2022) 104275, <http://dx.doi.org/10.1016/j.autcon.2022.104275>.
- [68] Y. Zhou, A. Ji, L. Zhang, Sewer defect detection from 3D point clouds using a transformer-based deep learning model, *Autom. Constr.* 136 (2022) 104163, <http://dx.doi.org/10.1016/j.autcon.2022.104163>.
- [69] S. Moon, S. Chi, S.-B. Im, Automated detection of contractual risk clauses from construction specifications using bidirectional encoder representations from transformers (BERT), *Autom. Constr.* 142 (2022) 104465, <http://dx.doi.org/10.1016/j.autcon.2022.104465>.
- [70] J. Devlin, M.-W. Chang, K. Lee, K. Toutanova, BERT: Pre-training of deep bidirectional transformers for language understanding, 2018, <http://dx.doi.org/10.48550/arXiv.1810.04805>, arXiv preprint [arXiv:1810.04805](http://dx.doi.org/10.48550/arXiv.1810.04805).
- [71] Z. Liu, Y. Lin, Y. Cao, H. Hu, Y. Wei, Z. Zhang, S. Lin, B. Guo, Swin transformer: Hierarchical vision transformer using shifted windows, in: *Proceedings of the IEEE International Conference on Computer Vision*, 2021, pp. 9992–10002, <http://dx.doi.org/10.1109/ICCV48922.2021.00986>.
- [72] J.L. Ba, J.R. Kiros, G.E. Hinton, Layer normalization, 2016, <http://dx.doi.org/10.48550/arXiv.1607.06450>, arXiv preprint [arXiv:1607.06450](http://dx.doi.org/10.48550/arXiv.1607.06450).
- [73] J. Hu, L. Shen, G. Sun, Squeeze-and-excitation networks, in: *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 2018, pp. 7132–7141, <http://dx.doi.org/10.1109/CVPR.2018.00745>.
- [74] D. Meng, X. Peng, K. Wang, Y. Qiao, Frame attention networks for facial expression recognition in videos, in: *Proceedings of the IEEE International Conference on Image Processing*, 2019, pp. 3866–3870, <http://dx.doi.org/10.1109/ICIP.2019.8803603>.
- [75] S.N. Gowda, M. Rohrbach, L. Sevilla-Lara, Smart frame selection for action recognition, 2020, <http://dx.doi.org/10.48550/arXiv.2012.10671>, arXiv preprint [arXiv:2012.10671](http://dx.doi.org/10.48550/arXiv.2012.10671).
- [76] F. Sung, Y. Yang, L. Zhang, T. Xiang, P.H. Torr, T.M. Hospedales, Learning to compare: Relation network for few-shot learning, in: *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 2018, pp. 1199–1208, <http://dx.doi.org/10.1109/CVPR.2018.00131>.
- [77] J. Ke, M. Zhang, X. Luo, J. Chen, Monitoring distraction of construction workers caused by noise using a wearable electroencephalography (EEG) device, *Autom. Constr.* 125 (2021) 103598, <http://dx.doi.org/10.1016/j.autcon.2021.103598>.
- [78] H. Lee, G. Lee, S. Lee, C.R. Ahn, Assessing exposure to slip, trip, and fall hazards based on abnormal gait patterns predicted from confidence interval estimation, *Autom. Constr.* 139 (2022) 104253, <http://dx.doi.org/10.1016/j.autcon.2022.104253>.
- [79] M. Koshkina, H. Pidaparthi, J.H. Elder, Contrastive learning for sports video: Unsupervised player classification, in: *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition Workshops*, 2021, pp. 4523–4531, <http://dx.doi.org/10.1109/CVPRW53098.2021.00510>.
- [80] R. Hadsell, S. Chopra, Y. LeCun, Dimensionality reduction by learning an invariant mapping, in: *IEEE Computer Society Conference on Computer Vision and Pattern Recognition*, 2006, pp. 1735–1742, <http://dx.doi.org/10.1109/CVPR.2006.100>.
- [81] K. He, X. Chen, S. Xie, Y. Li, P. Dollár, R. Girshick, Masked autoencoders are scalable vision learners, in: *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 2022, pp. 16000–16009, <http://dx.doi.org/10.48550/arXiv.2111.06377>.

- [82] K. Soomro, A.R. Zamir, M. Shah, UCF101: A dataset of 101 human actions classes from videos in the wild, 2012, <http://dx.doi.org/10.48550/arXiv.1212.0402>, arXiv preprint [arXiv:1212.0402](https://arxiv.org/abs/1212.0402).
- [83] R. Wightman, Pytorch image models, 2019, <http://dx.doi.org/10.5281/zenodo.4414861>, <https://github.com/rwightman/pytorch-image-models>.
- [84] K. Hara, H. Kataoka, Y. Satoh, Can spatiotemporal 3D CNNs retrace the history of 2D CNNs and ImageNet? in: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, 2018, pp. 6546–6555, <http://dx.doi.org/10.1109/CVPR.2018.00685>.
- [85] MMAAction2 Contributors, OpenMMLab's next generation video understanding toolbox and benchmark, 2020, <https://github.com/open-mmlab/mmaaction2>.
- [86] H. Fan, T. Murrell, H. Wang, K.V. Alwala, Y. Li, Y. Li, B. Xiong, N. Ravi, M. Li, H. Yang, J. Malik, R. Girshick, M. Feiszli, A. Adcock, W.-Y. Lo, C. Feichtenhofer, PyTorchVideo: A deep learning library for video understanding, in: Proceedings of the 29th ACM International Conference on Multimedia, 2021, pp. 3783–3786, <https://pytorchvideo.org/>.
- [87] T. Ridnik, E. Ben-Baruch, A. Noy, L. Zelnik-Manor, Imagenet-21k pretraining for the masses, 2021, <http://dx.doi.org/10.48550/arXiv.2104.10972>, arXiv preprint [arXiv:2104.10972](https://arxiv.org/abs/2104.10972).
- [88] W. Kay, J. Carreira, K. Simonyan, B. Zhang, C. Hillier, S. Vijayanarasimhan, F. Viola, T. Green, T. Back, P. Natsev, et al., The kinetics human action video dataset, 2017, <http://dx.doi.org/10.48550/arXiv.1705.06950>, arXiv preprint [arXiv:1705.06950](https://arxiv.org/abs/1705.06950).
- [89] I. Loshchilov, F. Hutter, Decoupled weight decay regularization, 2017, <http://dx.doi.org/10.48550/arXiv.1711.05101>, arXiv preprint [arXiv:1711.05101](https://arxiv.org/abs/1711.05101).
- [90] J. Deng, W. Dong, R. Socher, L.-J. Li, K. Li, L. Fei-Fei, ImageNet: A large-scale hierarchical image database, in: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, 2009, pp. 248–255, <http://dx.doi.org/10.1109/CVPR.2009.5206848>.
- [91] pytorch OpCounter Contributors, pytorch OpCounter, 2018, <https://github.com/Lyken17/pytorch-OpCounter>.
- [92] K. He, X. Zhang, S. Ren, J. Sun, Deep residual learning for image recognition, in: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, 2016, pp. 770–778, <http://dx.doi.org/10.1109/CVPR.2016.90>.
- [93] G. Huang, Z. Liu, L. Van Der Maaten, K.Q. Weinberger, Densely connected convolutional networks, in: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, 2017, pp. 2261–2269, <http://dx.doi.org/10.1109/CVPR.2017.243>.
- [94] Z. Teed, J. Deng, Raft: Recurrent all-pairs field transforms for optical flow, in: Proceedings of the European Conference on Computer Vision, Springer, 2020, pp. 402–419, <http://dx.doi.org/10.48550/arXiv.2003.12039>.